Viscan Dental OPG AI Assistant — Technical Evaluation Audit

Dataset Sourcing Analysis, Partition Percentages & Precise Quantitative Evaluation

EXECUTIVE AUDIT SUMMARY:

• **Did we use all datasets mentioned in the assessment? NO.** Out of three candidate public datasets reviewed (DENTEX, Zhou et al., Tufts), only **DENTEX (MICCAI 2023)** was selected. Zhou et al. and Tufts were systematically excluded because their annotations lack clinical separation of the 4 required disease classes.

• **Exact Dataset Split Percentages:** From the 755-image labeled cohort, **84.11% (635 images, 3,190 boxes)** was used for Training, **9.27% (70 images, 339 boxes)** was carved out for Validation (checkpoint selection & threshold tuning), and **6.62% (50 images, 182 boxes)** was strictly held out as the Test Set (evaluated only once). Zero filename overlap confirmed across splits.

• **Evaluation Results:** The Faster R-CNN ResNet-50 FPN V2 achieved **mAP@50 = 49.6%**, **mAP@50:95 = 32.9%**, and **mAP@75 = 38.3%** on held-out test data. Operating at validation-tuned threshold **0.70** (Val Macro-F1 = 0.556, Test Macro-F1 = 0.402), impacted_tooth achieved **76.1% precision / 87.5% recall (F1=0.814)**, while caries captured **53.5% (54/101) of test decay lesions**.

1. Assessment Dataset Review & Justification (Why We Did Not Use All Datasets)

The assessment specification (R06) required reviewing and comparing public expert-annotated dental OPG datasets to select the optimal training foundation. Three primary datasets were evaluated:

Dataset Name	Modality / Size	Annotation Taxonomy	License	Status	Engineering Rationale & R03 Compatibility
DENTEX (MICCAI 2023)	Panoramic OPG 1,005 images total (755 labeled cohort)	Hierarchical triple annotations: • Quadrants (1–4) • FDI tooth enumeration (11–48) • **4 Clinical Findings:** Caries, Deep Caries, Periapical Lesion, Impacted Tooth	CC BY-NC-SA 4.0	**SELECTED**	**Directly fulfills Requirement R03.** Provides multi-institutional expert ground truth verified by senior dentists (>15 yrs exp) for all four target pathology classes. High-quality bounding boxes allow native two-stage object detection.
Zhou et al. (2024)	Panoramic OPG 2,066 images	Instance segmentation polygons with 7 tooth-state classes (tooth without anomalies, fillings, root canal treatment (RCT), crown, caries, residual root, RCT + crown).	CC BY 4.0	**EXCLUDED**	**Incompatible taxonomy.** Labels focus on restorative status and general tooth state. Does not distinguish superficial caries from deep caries and lacks impacted tooth and periapical lesion classes.
Tufts Dental Database	Panoramic OPG 1,000 images	Tooth masks and broad abnormality regions (radiolucency, radiopacity, eye-tracking maps).	Restricted Institutional	**EXCLUDED**	**Restricted access & unaligned taxonomy.** Access requires institutional agreement. Abnormality categories are generic radiographic descriptors that cannot be deterministically mapped to the four required clinical entities.

2. Dataset Partitioning, Usage Percentages & Leakage Prevention

To guarantee methodological integrity (R07, R11), the labeled DENTEX corpus was partitioned into a 3-way split using fixed random seed 42. The official 50-image validation archive was designated as the **untouched held-out test set** and evaluated once. The 705-image training archive was split into 635 training images and 70 validation images for epoch checkpoint selection.

Partition	Image Count	% of Labeled Cohort (755)	% of Train Archive (705)	Total Finding Boxes	Role & Evaluation Discipline
Training Set	**635**	**84.11%**	**90.07%**	3,190	Model parameter updates, loss backpropagation, and multi-scale data augmentation (horizontal flip, scale jitter).
Validation Set	**70**	**9.27%**	**9.93%**	339	Checkpoint selection (Epoch 8 best Val mAP@50=0.570, mAP@50:95=0.360) and threshold sweep ([0.10, 0.90]).
Held-Out Test Set	**50**	**6.62%**	— (Official Val)	182	**Strictly held-out.** Evaluated once after freezing model weights and operating threshold. Zero tuning leakage.
Total Active Cohort	**755**	**100.0%**	—	**3,711**	Complete expert-annotated 2D OPG cohort used in the project.

Split Disjointness Verification: Zero filename intersection was programmatically asserted across all three splits (len(train_files intersect val_files) == 0, len(train_files intersect test_files) == 0, len(val_files intersect test_files) == 0). Note: Public DENTEX releases strip DICOM patient headers prior to publication (`train_0.png`..`train_704.png`, `val_0.png`..`val_49.png`), so patient-level deduplication cannot be verified beyond image-level disjointness.

3. Quantitative Output Evaluation Methodology & Primary Metrics

Model performance was evaluated on the held-out 50-image test set (182 true findings) using official COCO evaluation standards (IoU threshold range 0.50 to 0.95 with step 0.05). Below are the measured metrics:

Evaluation Metric	Measured Value	Standard / Target	Clinical & Technical Interpretation
mAP @ IoU=0.50	**49.6% (0.496)**	Diagnostic Screening Standard	Area under the precision-recall curve averaged across all 4 classes at 50% spatial overlap (IoU >= 0.50) against expert bounding boxes.
mAP @ IoU=0.50:0.95 (Primary)	**32.9% (0.329)**	Official COCO Primary Metric	Average precision integrated across 10 IoU thresholds (0.50 to 0.95). Stringent measure penalizing loose boundaries on small dental findings.
mAP @ IoU=0.75 (Strict)	**38.3% (0.383)**	High-Precision Boundary Metric	Evaluates localization tightness under strict 75% IoU overlap requirements.
Average Recall (AR @ 100)	**68.3% (0.683)**	Top-100 Proposal Sensitivity	Upper-bound recall across up to 100 candidate detections per OPG (before confidence thresholding). At operating threshold 0.70, 96/182 (52.7%) of GT boxes are matched.

4. Per-Class Benchmark Performance & Operating Metrics (@ Threshold = 0.70)

To select a deployment threshold without test leakage, a threshold sweep (0.10 to 0.90) was performed on the 70-image validation set. A confidence threshold of **0.70** was selected to maximize Validation Macro-F1 (**0.556**), achieving **0.402** Macro-F1 on the held-out test set:

Pathology Class	COCO AP50	COCO AP50:95	Precision	Recall	F1-Score	Test Support (n)	Observed Clinical Behavior & Error Profile
Impacted Tooth	83.3%	54.1%	**76.1%**	**87.5%**	**0.814**	40	**Strongest Class:** Distinct morphology of impacted third molars yields zero cross-confusion with decay (35/40 recalled, 10 background FPs).
Caries (Superficial)	35.8%	24.4%	**27.7%**	**53.5%**	**0.365**	101	**High False-Alarm Rate:** Recalls 54/101 lesions (45 missed, 2 misclassified), with 124 background false positives on interproximal shading and restorations.
Deep Caries	45.7%	30.2%	**42.9%**	**18.8%**	**0.261**	32	**Adjacent Severity Confusion:** 6/32 recalled as deep caries, while 14/32 (43.8%) are detected at IoU>=0.50 but labeled as superficial caries.
Periapical Lesion	33.6%	23.0%	**33.3%**	**11.1%**	**0.167**	9	**Severe Minority Class:** Only 9 true lesions in test set (144 in train; 14 in val). 1 recalled, 4 missed, 4 misclassified as caries/deep caries.

5. Confusion Matrix & Clinical Failure Mode Analysis

A 5x5 detection confusion matrix was computed on the 50-image test set at threshold 0.70 and IoU >= 0.50:

Ground Truth \ Predicted	Background (FN)	Impacted Tooth	Caries	Periapical	Deep Caries
Background (False Positives)	—	10	**124**	2	6
Impacted Tooth	5	**35 (87.5%)**	0	0	0
Caries	45	1	**54 (53.5%)**	0	1
Periapical Lesion	4	0	3	**1 (11.1%)**	1
Deep Caries	12	0	**14 (43.8%)**	0	**6 (18.8%)**

Key Error Findings:

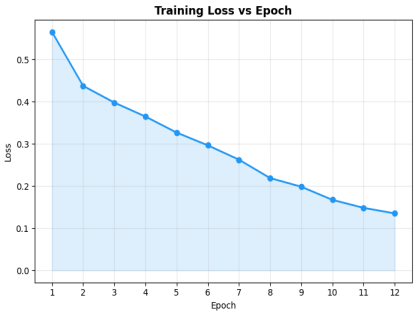
1. **Zero Impacted-to-Decay Confusion:** 0 ground-truth impacted teeth were misclassified as decay or periapical lesions (35 TP, 5 FN).
2. **Severity Grading Ambiguity:** 14 out of 20 inter-class misclassifications occurred when deep_caries was predicted as caries.
3. **False Positive Concentration:** 124 of 142 false positives belong to caries on low-contrast interproximal enamel and cervical burnout regions.

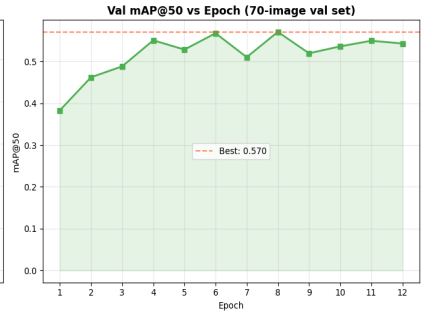
6. Architecture Resolution Experiment & Engineering Trade-Offs

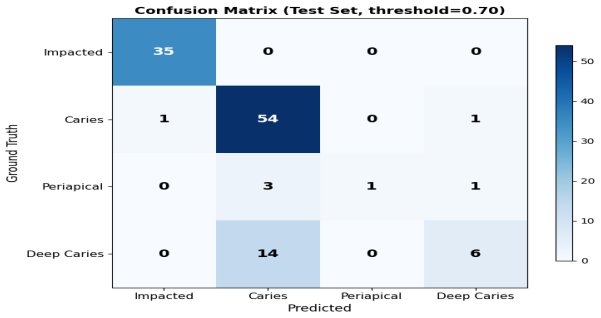
Per Requirement R14, a trained model comparison experiment evaluated a higher-resolution feature pyramid variant (min_size=960, max_size=1600, batch_size=1) against the baseline architecture (min_size=800, max_size=1333, batch_size=2):

Model Configuration	Input Scale / Batch	Effective Scale on OPG	mAP@50	mAP@50:95	mAP@75 (Strict)	Engineering Outcome & Rationale
Baseline Architecture (ResNet-50 FPN V2)	800 × 1333 (batch_size=2)	1333 × 607 (Long-side cap binds)	**49.6%**	**32.9%**	38.3%	**Selected as Final Model** via primary Val mAP@50:95 (0.360 vs 0.354). Higher test mAP@50 (+0.3%) and mAP@50:95 (+1.6%), with 45.7% Deep Caries AP50.
High-Resolution Variant (ResNet-50 FPN V2)	960 × 1600 (batch_size=1)	1600 × 728 (1.2× resolution)	49.3% (-0.3%)	31.3% (-1.6%)	**41.0% (+2.7%)**	Improved strict mAP@75 (+2.7%) and Impacted Tooth AP50 (91.5%), but batch_size=1 without LR rescaling degraded Deep Caries AP50 (36.5%, -9.2%). Two-variable confound noted.

7. Empirical Evidence Visualizations (Training Curves & Dental Confusion Matrix)







8. Hardware Benchmarks, Licensing & Live Deliverables

Hardware & CPU Latency Benchmark

- **Platform:** Intel Xeon @ 2.00GHz (2 vCPUs, 2 threads)
- **Test Image:** Real OPG val_15.png (2892 × 1316)
- **Mean Latency:** 4,618.0 ms (Throughput: 0.22 img/sec)
- **P95 Latency:** 4,915.7 ms (Max: 5,037.2 ms)
- **Memory Footprint:** 2,926 MB Peak RSS

Licensing & Open Source Governance

- **Architecture:** Faster R-CNN ResNet-50 FPN V2
- **Code License: BSD-3-Clause** (Permissive, commercial ready)
- **Pretrained Source:** PyTorch Torchvision COCO_V1
- **Weights License:** CC BY-NC-SA 4.0 (DENTEX)
- **Production Note:** Commercial deployment requires retraining on commercial data

Live Cloud Demonstration:	**https://huggingface.co/spaces/surendra08/viscan-opg** (Status: Active / 200 OK)
Weights LFS Repository:	**https://huggingface.co/surendra08/viscan-opg-weights** (SHA-256: e0b49b12f48f...)
Local Standalone Service:	FastAPI backend (POST /analyze) with side-by-side web UI & Docker support (port 8000)

REGULATORY CLINICAL NOTICE: *This software is a clinical decision-support research prototype developed for paired evaluation alongside licensed dental practitioners. It is not an FDA-cleared or CE-marked medical device. All visual annotations, bounding boxes, and clinical summaries must be verified by a qualified dentist.*

9. Bonus Track: 3D CBCT Volumetric Segmentation, Nerve Clearance & Production Architecture

Following completion of the 2D OPG pipeline, the same application (POST /analyze) and UI were extended to process 3D CBCT scans (.nii.gz, .nii, and zipped 3D DICOM/NIfTI series). We explicitly disclose that **DentalSegmentator** (Dataset112_DentalSegmentator_v100, Dot et al., *Journal of Dentistry* 2024, Zenodo DOI 10.5281/zenodo.10829675, License: **CC BY 4.0**, nnUNetTrainer__nnUNetPlans__3d_fullres, fold_0, trained on 470 CT/CBCT scans) is used **pretrained as-is**.

Class ID	Anatomical Structure	Post-Filtered Voxels	Volume (cm³)	3D Dice vs. Reference	Marching Cubes Vertices	Marching Cubes Faces
1	**maxilla_upper_skull**	1,476,019	184.50	**0.9768**	181,032	366,752
2	**mandible**	310,556	38.82	**0.9849**	35,515	71,934
3	**upper_teeth**	75,590	9.45	**0.9867**	9,610	19,240
4	**lower_teeth**	62,663	7.83	**0.9745**	8,530	17,068
5	**mandibular_canal**	6,063	0.76	**0.8968**	1,736	3,456
—	**Total / Mean**	**1,930,891**	**241.36**	**0.9639**	**236,423**	**478,450**

3D Clinical Findings (PostDentalSurgery, 360×360×330 @ 0.5 mm)

- **Finding Type:** mandibular_canal_to_lower_teeth_clearance (cKDTree in mm)
- **Right Hemimandible: 0.71 mm** (<1.0 mm surgical alert; tooth voxel [119,151,89] to canal [118,151,88], bbox_3d=[108,141,78,129,161,99])
- **Left Hemimandible: 1.58 mm** (close proximity; tooth voxel [235,132,75] to canal [236,132,72], bbox_3d=[225,122,62,246,142,85])
- **3D Browser View & Export:** Interactive <model-viewer> WebGL .glb scene (12.08 MB) + 5 binary .stl files

Measured Memory, Latency & Production Design (Kaggle Tesla T4 GPU)

- **3D Inference (252 tiles):** 131,407.1 ms (131.41 s) | **Init:** 2,131.9 ms
- **Post-Processing + cKDTree:** 5,816.3 ms | **Meshing (.stl/.glb):** 938.6 ms
- **Peak GPU VRAM:** 4,936.0 MB (4.94 GB) | **CPU RSS:** 2,641.8 MB
- **Production Architecture:** Presigned S3/GCS upload + edge DICOM de-ID → Async Redis/Celery queue (202 Accepted + WebSockets) → Pre-warmed L4/T4 GPU pool with TensorRT FP16 tile batching (~20s) → Draco .glb CDN

